

DDA3020 Machine Learning

Tutorial: Performance Evaluation

April 2026

Instructions

This tutorial is based on Lecture 13 *Performance Evaluation*. Its main focuses contains the key concepts emphasized in the lecture slides: motivation for performance evaluation, hyper-parameter tuning, K-fold cross-validation, regression metrics, binary-classification metrics, confusion matrix, threshold-dependent behavior, DET/ROC curves, AUC and multcategory confusion matrix.

Useful Formulas

- **Regression metrics**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|.$$

- **Binary classification metrics**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

- **Cost-sensitive accuracy**

$$\frac{C_{p,p}TP + C_{n,n}TN}{C_{p,p}TP + C_{n,n}TN + C_{p,n}FN + C_{n,p}FP}.$$

- **Normalized rates**

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FNR} = \frac{FN}{TP + FN}, \quad \text{TNR} = \frac{TN}{FP + TN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

$$\text{TPR} + \text{FNR} = 1, \quad \text{TNR} + \text{FPR} = 1.$$

- **Balanced-class identity**

$$\text{Accuracy} = \frac{\text{TPR} + \text{TNR}}{2} = 1 - \frac{\text{FPR} + \text{FNR}}{2}.$$

- **Equal Error Rate (EER):** The intersection point where:

$$\text{FPR} = \text{FNR}.$$

- **AUC formula**

$$e_{ij} = g(x_i^+) - g(x_j^-), \quad \text{AUC} = \frac{1}{m_+ m_-} \sum_{i=1}^{m_+} \sum_{j=1}^{m_-} u(e_{ij}).$$

where

$$u(e) = \begin{cases} 1, & e > 0, \\ 0.5, & e = 0, \\ 0, & e < 0. \end{cases}$$

Definition Review

1. Motivation We evaluate a learning algorithm because we want to know how well it will perform on *unseen data*, using only the limited data we currently have.

2. Cross-validation

2.1. Hyper-parameter tuning: Hyper-parameters are settings chosen *outside* training, while parameters are learned from data during training. Examples of **Hyper-parameter** are as follow:

- Polynomial model: polynomial order.
- Decision tree: maximum depth.
- Random forest: number of trees.
- SVM: trade-off parameter C .
- Gradient descent: learning rate.

2.2. K -fold cross-validation: Its main purpose is to choose hyper-parameters more reliably than using only one random train/validation split.

- Split the training data into K folds.
- Use one fold for validation and the remaining $K - 1$ folds for training.
- Repeat until every fold has served once as the validation set.
- Average the validation results and choose the best hyper-parameter setting.

In the lecture slide, $K = 5$ to 10 is stated as common in practice.

3. Evaluation metrics for regression: The lecture emphasizes two standard metrics: **MSE**, which penalizes large errors more heavily, and **MAE**, which measures the average absolute prediction error.

4. Evaluation metrics for classification

4.1. Confusion matrix: For binary classification, class 1 is treated as the positive class and class 2 as the negative class. The four entries are TP, TN, FP, and FN.

4.2. Accuracy, precision, and recall: Accuracy measures overall correctness. Precision tells us how reliable the positive predictions are. Recall tells us how many actual positive samples are successfully found.

4.3. Threshold / operating point: Changing the decision threshold changes the confusion matrix, so it also changes TPR, FPR, FNR, and TNR. Each threshold corresponds to one operating point.

4.4. EER: As the threshold changes, FPR usually decreases while FNR usually increases. Their intersection is the Equal Error Rate (EER).

4.5. DET curve, ROC curve, and AUC: The DET curve plots FNR against FPR. The ROC curve plots TPR against FPR. AUC summarizes overall ranking/classification performance across all possible thresholds.

- $\text{AUC} = 1$: perfect ranking.
- $\text{AUC} = 0.5$: random guessing.
- $\text{AUC} = 0$: completely reversed ranking.

5. Multicategory classification: The confusion matrix can still be defined for multiple classes, but multiclass ROC is much more complicated and has no consensus definition.

6. Computational efficiency vs. maintainability (optional): Higher computational efficiency does not always come together with better maintainability, flexibility, or portability, so this trade-off should be considered carefully.

Exercises

Q1. In Tom Mitchell's definition of machine learning, what do *experience* E , *task* T , and *performance measure* P correspond to in supervised learning?

Q2. The lecture states that a learning algorithm we have seen so far contains the following three parts:

- a loss function,
- an objective function / optimization criterion,
- an optimization routine.

For each part, briefly explain its role in the learning process. Also explain why the lecture says this is only *one part of the workflow of machine learning*, rather than the whole workflow.

Q3. Explain the model selection problem using the bias–variance trade-off discussed in the lecture. Why can both a very simple model and a very complex model lead to high test prediction error?

Q4. Which of the following quantities is a *hyper-parameter* rather than a learned parameter?

- (A) The weight vector w in linear regression
- (B) The bias term b in logistic regression
- (C) The maximum tree depth in a decision tree
- (D) The fitted coefficients of a Lasso model

Q5. The lecture presents four ideas for tuning hyper-parameters.

- (a) Why is it a bad idea to choose hyper-parameters that work best on the whole dataset directly?
- (b) Why is it also a bad idea to choose hyper-parameters on the test set?
- (c) Why is train/validation/test splitting better, but still not fully satisfactory?

Q6. Describe the basic procedure of K -fold cross-validation in four clear steps.

Q7. According to the lecture, what happens if K is too large or too small in K -fold cross-validation? What range of K is commonly used in practice?

Q8. Suppose a regression model is evaluated on 4 samples with true outputs

$$(y_1, y_2, y_3, y_4) = (2, 0, 3, 5)$$

and predicted outputs

$$(\hat{y}_1, \hat{y}_2, \hat{y}_3, \hat{y}_4) = (1, 1, 4, 3).$$

Compute:

- (a) MSE
- (b) MAE

Q9. Consider a binary classifier with the following confusion matrix entries:

$$TP = 18, \quad TN = 30, \quad FP = 6, \quad FN = 6.$$

Compute:

- (a) Accuracy
- (b) Precision
- (c) Recall
- (d) TPR, TNR, FPR, and FNR

Q10. In a medical diagnosis task, false negatives are much more costly than false positives. Based on the lecture, explain why plain accuracy may not be a good metric in this case. What metric introduced in the lecture can better reflect this situation?

Q11. The lecture emphasizes that changing the decision threshold changes the confusion matrix. Explain what an *operating point* is and why different thresholds lead to different FPR/FNR values.

Q12. The lecture introduces both the DET curve and the ROC curve.

- (a) State what is plotted on the x-axis and y-axis of each curve.
- (b) Explain what it means visually if a DET curve bends more toward the bottom-left corner.
- (c) Explain what it means visually if a ROC curve bends more toward the top-left corner.
- (d) The lecture also mentions the Equal Error Rate (EER). On the threshold-based error plot, what does the EER point represent?

Q13. Which of the following statements about ROC and AUC is correct?

- (A) AUC depends heavily on the numerical scale of the prediction scores.
- (B) AUC changes whenever the classification threshold changes.
- (C) A higher ROC curve bending toward the top-left corner indicates better classification performance.

(D) AUC can only take values between 0.5 and 1.

Q14. Use the AUC formula from the lecture. Suppose there are two positive samples and two negative samples with prediction scores

$$g(x_1^-) = 0.2, \quad g(x_2^-) = 0.7, \quad g(x_1^+) = 0.5, \quad g(x_2^+) = 0.9.$$

Compute the AUC.

Q15. The lecture presents four typical AUC cases. Briefly interpret the following values:

- (a) AUC = 1
- (b) AUC = 0.5
- (c) AUC = 0

Solutions

Q1. Solution. In supervised learning:

- experience E corresponds to the **training data**,
- task T corresponds to **classification or regression**,
- performance measure P corresponds to the **evaluation metric** used to judge how well the algorithm performs.

Q2. Solution. The three parts play different roles:

- A **loss function** defines how wrong the prediction is on the training data.
- An **objective function / optimization criterion** specifies what quantity we want to minimize or optimize based on that loss.
- An **optimization routine** is the algorithmic procedure used to find a solution, for example gradient descent.

The lecture also emphasizes that this is only **one part of the workflow of machine learning**. In practice, machine learning also includes evaluation and model selection, because after training we still need to assess how well the learned model performs on novel data.

Q3. Solution. A model with very low complexity tends to have **high bias** and **low variance**, which corresponds to **underfitting**; therefore its test error can be high. A model with very high complexity tends to have **low bias** but **high variance**, which corresponds to **overfitting**; its test error can also be high. The model selection problem is to find a suitable complexity that balances these two effects.

Q4. Solution. Answer: C.

The maximum tree depth is chosen outside the learning procedure, so it is a hyper-parameter. The others are parameters learned from data.

Q5. Solution.

- (a) Choosing hyper-parameters that work best on the whole dataset can make the system look perfect on the available data, but it may generalize poorly to novel data.

- (b) Choosing hyper-parameters on the test set is also bad because then the test set is no longer truly unseen; we lose a reliable estimate of performance on new data.
- (c) Splitting into train/validation/test is better because the test set is not involved in training or tuning. However, the result can still depend heavily on one random split of the train and validation sets.

Q6. Solution. A standard K -fold cross-validation procedure is:

1. Split the training data into K folds.
2. Use one fold as validation and the remaining $K - 1$ folds as training.
3. Train the model on the training folds and evaluate it on the validation fold.
4. Repeat this for all folds, average the validation results, and choose the hyper-parameters with the best average result.

Q7. Solution. If K is too large, the validation set becomes too small and the training sets across trials become too similar, which may cause overfitting. If K is too small, the training data in each trial may be insufficient, which may cause underfitting. In practice, the lecture suggests using $K = 5$ to 10.

Q8. Solution. The errors are

$$(2 - 1, 0 - 1, 3 - 4, 5 - 3) = (1, -1, -1, 2).$$

Squared errors are

$$1, 1, 1, 4,$$

so

$$\text{MSE} = \frac{1 + 1 + 1 + 4}{4} = \frac{7}{4} = 1.75.$$

Absolute errors are

$$1, 1, 1, 2,$$

so

$$\text{MAE} = \frac{1 + 1 + 1 + 2}{4} = \frac{5}{4} = 1.25.$$

Q9. Solution.

(a)

$$\text{Accuracy} = \frac{18 + 30}{18 + 30 + 6 + 6} = \frac{48}{60} = 0.8.$$

(b)

$$\text{Precision} = \frac{18}{18 + 6} = \frac{18}{24} = 0.75.$$

(c)

$$\text{Recall} = \frac{18}{18 + 6} = \frac{18}{24} = 0.75.$$

(d)

$$\begin{aligned} \text{TPR} &= \frac{18}{18 + 6} = 0.75, & \text{FNR} &= \frac{6}{18 + 6} = 0.25, \\ \text{TNR} &= \frac{30}{30 + 6} = \frac{30}{36} = \frac{5}{6} \approx 0.8333, & \text{FPR} &= \frac{6}{30 + 6} = \frac{1}{6} \approx 0.1667. \end{aligned}$$

Q10. Solution. If false negatives are much more costly than false positives, two classifiers with similar accuracy may still have very different practical usefulness. Accuracy treats all mistakes equally, so it may hide the real cost of dangerous errors. The lecture proposes using **cost-sensitive accuracy**, where different costs are assigned to different kinds of mistakes.

Q11. Solution. An **operating point** is the pair of error-rate values obtained at a particular decision threshold. When we move the threshold, some samples switch from predicted positive to predicted negative, or vice versa. As a result, TP, TN, FP, and FN change, so FPR and FNR also change.

Q12. Solution.

- (a) For the **DET curve**, the x-axis is **FPR** and the y-axis is **FNR**. For the **ROC curve**, the x-axis is **FPR** and the y-axis is **TPR**.
- (b) If a DET curve bends more toward the **bottom-left corner**, it means both FPR and FNR are relatively small, so the classifier has better performance.
- (c) If a ROC curve bends more toward the **top-left corner**, it means the classifier achieves high TPR while keeping FPR low, which indicates better performance.
- (d) The **EER** point is the point where the two error rates are equal, namely **FPR = FNR** on the threshold-based error plot.

Q13. Solution. Answer: C.

A ROC curve closer to the top-left corner indicates better performance. Option A is false because AUC is scale-invariant. Option B is false because AUC is threshold-invariant. Option D is false because the lecture explicitly discusses AUC values from 0 to 1.

Q14. Solution. There are $m_+ = 2$ positive samples and $m_- = 2$ negative samples, so we need 4 pairwise comparisons.

$$e_{11} = g(x_1^+) - g(x_1^-) = 0.5 - 0.2 = 0.3,$$

$$e_{12} = g(x_1^+) - g(x_2^-) = 0.5 - 0.7 = -0.2,$$

$$e_{21} = g(x_2^+) - g(x_1^-) = 0.9 - 0.2 = 0.7,$$

$$e_{22} = g(x_2^+) - g(x_2^-) = 0.9 - 0.7 = 0.2.$$

Applying $u(e)$,

$$u(e_{11}) = 1, \quad u(e_{12}) = 0, \quad u(e_{21}) = 1, \quad u(e_{22}) = 1.$$

Therefore,

$$\text{AUC} = \frac{1 + 0 + 1 + 1}{4} = \frac{3}{4} = 0.75.$$

Q15. Solution.

- (a) $\text{AUC} = 1$: perfect ranking; every positive sample is ranked above every negative sample.
- (b) $\text{AUC} = 0.5$: no better than random guessing.
- (c) $\text{AUC} = 0$: completely reversed ranking; all positive samples are ranked below all negative samples.